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Express $\cos(3x)$ as a function of $\cos(x)$:

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\begin{equation*}
\cos(3x)=f(\cos(x));
\end{equation*}
```

then find $\cos(x)$ by solving

```
\begin{equation*}
\cos(3x)-f(\cos(x))=0
\end{equation*}
```

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First use the triple-angle identity:

$$\cos(3x) = 4 \cos^3 x - 3 \cos x.$$

So the function is

$$f(u) = 4u^3 - 3u,$$

and therefore

$$\cos(3x) = f(\cos x) = 4 \cos^3 x - 3 \cos x.$$

Now consider

$$\cos(3x) - f(\cos x) = 0.$$

Substituting $f(\cos x)$ gives

$$\cos(3x) - (4 \cos^3 x - 3 \cos x) = 0.$$

But since $\cos(3x)$ is *exactly equal* to $4 \cos^3 x - 3 \cos x$ for all real x , the left-hand side becomes

$$0 = 0.$$

Conclusion

- The identity holds for **all real** x .
- Hence there is **no restriction on** $\cos x$ beyond the usual range:

$$\cos x \in [-1, 1].$$

So the equation does not determine specific values of $\cos x$; it is an identity rather than a solvable constraint.



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